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Saveetha Institute of Medical And Technical Sciences

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Cloud Based Smart Warehouse Automation and Inventory Analytics Using Big Data.

CSA 1501:Cloud Computing and big data analytics

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ABSTRACT

- The system provides cloud-based management for monitoring inventory across multiple warehouses in real time.
- It uses big data analytics and AI to forecast product demand and optimize inventory levels.
- The platform automatically calculates reorder points to reduce stock shortages and overstocking.
- An explainable anomaly detection model identifies unusual inventory activities and provides the reason behind each alert.
- The system optimizes stock transfers between warehouses to improve efficiency and reduce transportation costs.
- It includes advanced security features, such as access-anomaly detection and a tamper-evident audit ledger, to ensure data integrity.

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INTRODUCTION

- Cloud computing is transforming warehouse operations by providing scalable, real-time storage and processing for stock data across multiple locations.
- Manual inventory tracking causes stockouts, overstocking, and undetected shrinkage that hurt profitability and customer trust.
- A cloud-based warehouse system helps managers detect reorder needs and loss patterns before they become costly problems.
- Big data analytics and machine learning enable demand forecasting and anomaly detection across large volumes of stock-movement data.
- This project combines cloud storage, explainable AI, and cost-optimization techniques into one integrated warehouse automation platform.

OBJECTIVES

- To develop a cloud-based platform that tracks inventory across multiple warehouses in real time.
- To forecast demand and calculate reorder points using an explainable machine learning model.
- To detect and explain abnormal stock-out events that may indicate shrinkage, theft, or damage.
- To optimize cloud storage and compute costs while keeping the inventory system secure and auditable.

Module 1:Cloud Inventory Tracking and Demand Forecasting

- Stores daily stock-movement data (stock in/out, closing stock) for every warehouse and SKU in a cloud database layer.
- Forecasts 14-day demand using a trend-plus-seasonality model, with confidence bands that widen with the forecast horizon.
- Calculates the reorder point from lead-time demand plus safety stock, and flags items that need reordering.
- Adjusts forecasts for known calendar events such as festivals and paydays, with an explainable multiplier.
- Recommends inter-warehouse stock transfers when one warehouse has surplus and another is running low.

Module 2: Explainable Shrinkage Detection and Root-Cause Analytics

- Runs an Isolation Forest model to flag stock-out events that deviate sharply from normal demand patterns.
- Attaches a plain-language likely cause (shrinkage, damage, or data error) to every flagged event instead of a raw score.
- Clusters flagged events by warehouse, category, and weekday to surface investigation patterns, not isolated alerts.
- Ranks shrinkage events by estimated rupee value lost, so the costliest issues get attention first.
- Logs every alert to a hash-chained audit ledger so past records cannot be silently altered.

Module 3:- Cloud Cost Optimization and Security

- Simulates cost-aware cloud storage tiering, automatically moving old inventory logs to cheaper storage tiers.

- Simulates API auto-scaling with request load, comparing cost against a fixed-capacity baseline.

- Detects access anomalies on the inventory system itself, such as off-hours edits or logins from unusual locations.

- Answers plain-English manager queries and generates SMS/WhatsApp-style alert digests for critical events.

Module 4: Security & Alert Management

- **Access Anomaly Detection:** Detects unusual login times, locations, and inventory access activities.
- **Tamper-Evident Audit Ledger:** Uses a hash-chained ledger to protect records from unauthorized modification.
- **Alert Management:** Generates alerts for critical inventory and security events.
- **Manager Query System:** Allows managers to ask inventory-related queries in plain English.

PROJECT COMPONENT REQUIREMENTS

- **Hardware Requirements**
  - Intel Core i5 Processor or higher
  - 8 GB RAM
  - 256 GB SSD
  - Laptop/Desktop with Internet Connection
  - Cloud Storage (SQLite / AWS RDS)

- **Python Libraries**
  - Pandas, NumPy
  - Scikit-learn
  - FastAPI
  - Streamlit

- **Software Requirements**
  - Windows 10/11
  - Python
  - Visual Studio Code
  - FastAPI Framework
  - SQLite / Cloud DB

- **Dataset Requirements**
  - Synthetic Warehouse Inventory Dataset
  - Multi-Warehouse Stock-Movement Logs
  - Festival/Event Calendar Dataset

TASKS

- Generate a realistic multi-warehouse stock-movement dataset with seasonal demand and injected anomalies.
- Build the cloud database layer and load it with the processed inventory data.
- Train the demand-forecast and shrinkage-detection models, and validate every endpoint via the FastAPI backend.
- Build the Streamlit dashboard and wire in all eleven novelty modules end to end.

SPECIFIC PARAMETERS

- **Stock Level:** Current closing stock per item, tracked daily across every warehouse.
- **Product Demand:** Historical stock-out volume used as the basis for the forecast model.
- **Lead Time & Safety Stock:** Used to calculate each item's reorder point.
- **Inventory Accuracy:** Deviation between expected and actual stock-out, used to flag shrinkage events.
- **Reorder Efficiency:** Cost and time saved via inter-warehouse transfers versus fresh purchase orders.

OUTCOME PARAMETERS

- **Reorder Recommendation:** Displays which items need reordering with a plain-language explanation.
- **Shrinkage Risk Classification:** Classifies flagged events as likely theft, damage, or data error.
- **Transfer Suggestion:** Recommends inter-warehouse stock transfers to avoid a fresh purchase order.
- **Cost Savings Report:** Shows estimated savings from cloud storage tiering and auto-scaling.
- **Alert Digest:** Generates an SMS/WhatsApp-style daily alert summary for the warehouse manager.

TECHNIQUES TO BE USED

- **Cloud Computing:** SQLite/AWS RDS-style cloud database layer for multi-warehouse data storage.
- **Machine Learning:** Isolation Forest for shrinkage detection, KMeans for root-cause clustering.
- **Backend & Frontend:** FastAPI for the cloud API layer, Streamlit for the interactive dashboard.
- **Explainability Layer:** Rule-based reasoning attached to every forecast and shrinkage alert.
- **Security:** Hash-chained tamper-evident audit ledger and access-anomaly detection.

FUTURE SCOPE

- Integrate real IoT/RFID sensors for live stock-movement data instead of synthetic data.
- Deploy the system on live AWS/Azure infrastructure with real storage-tier pricing.
- Improve forecast accuracy using LSTM-based deep learning on larger historical datasets.
- Connect the natural-language query module to a real voice assistant for hands-free use.
- Develop a native mobile app for warehouse managers with push notifications for alerts.

CONCLUSION

- The platform tracks stock and forecasts demand accurately using explainable ML.
- It detects shrinkage events and explains the likely cause instead of a black-box alert.
- Eleven novelty modules extend the system across forecasting, security, and cost optimization.
- It helps warehouse managers make informed, cost-efficient decisions through timely, explainable alerts.
- Overall, the project delivers a practical, auditable, cost-aware cloud solution for warehouse automation

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Thank you